11/12/2024

Phase-1_Code-A

Aakash
Medical | IIT-JEE | Foundations

Corporate Office : Aakash Tower, 8, Pusa Road, New Delhi-110005, *Ph.*011-47623456

MM : 720 NCERT Booster Test Series for NEET (OYM)-2024-25_Test-02 **Time : 200 Min.**

Topics Covered:
Physics: Alternating Current, Electromagnetic Waves, Ray Optics & Optical Instruments, Wave Optics, Dual Nature of Radiation and Matter, Atoms
Chemistry: Chemical Kinetics, Haloalkanes and Haloarenes, Alcohols, Phenols and Ethers, Aldehydes, Ketones and Carboxylic Acids
Botany: Principles of Inheritance and Variation, Molecular Basis of Inheritance-I (Upto: Transcription)
Zoology: Human Reproduction, Reproductive Health

General Instructions :

1. There are two sections in each subject, i.e. **Section-A & Section-B**. You have to attempt all **35 questions** from Section-A & only **10 questions** from Section-B out of **15.**
2. Each question carries **+4 marks**. For every wrong response, **–1 mark** shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
3. Use blue/black ballpoint pen only to darken the appropriate circle.
4. Mark should be dark and completely fill the circle.
5. Dark only one circle for each entry.
6. Dark the circle in the space provided only.
7. Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

CC-108 CC-108 CC-108 CC-108 CC-108 CC-108 CC-108

PHYSICS

SECTION-A

1. The resonant frequency in a series *LCR* AC circuit is

(1) $\frac{1}{2\pi\sqrt{LC}}$

(2) $2\pi\sqrt{LC}$

(3) $\frac{2\pi}{\sqrt{LC}}$

(4) $\frac{1}{2\pi}\sqrt{\frac{L}{C}}$

2. Consider a series LCR circuit in which $L = \frac{2}{\pi}$ H, $C = \frac{100}{\pi}\mu$F and resistance R = 100 Ω. When circuit is connected with an a.c. source 220 V, 50 Hz. The current drawn from the source will be

(1) $\sqrt{2}$ A

(2) $\frac{11\sqrt{2}}{10}$A

(3) $\sqrt{5}$ A

(4) $\frac{11\sqrt{5}}{10}$A

3. Power factor of series LCR circuit at resonance is

(1) $\frac{1}{\sqrt{2}}$

(2) $\frac{\sqrt{3}}{2}$

(3) 1

(4) Zero

4. An electromagnetic wave is propagating in vacuum along *x*-axis. If the electric field component is given by $E_Z = 2E_0 \sin(kx - \omega t)$, then magnetic field component is given by

(1) $B_x = \frac{E_0}{c}\sin\left(kx - \omega t\right)$

(2) $B_z = \frac{2E_0}{c}\sin\left(kx - \omega t\right)$

(3) $B_y = 2E_0 \sin(kx - \omega t)$

(4) $B_y = -\frac{2E_0}{c}\sin\left(kx - \omega t\right)$

5. Velocity of light in a medium with electrical permittivity ε and magnetic permeability μ is given by

(1) $\sqrt{\frac{\mu}{\varepsilon}}$

(2) $\sqrt{\mu\varepsilon}$

(3) $\frac{1}{\sqrt{\mu\varepsilon}}$

(4) $\sqrt{\frac{\varepsilon}{\mu}}$

6. A capacitor having a capacity of 4 pF. Potential difference across the capacitor is changing with a value of 10^{12} V s^{-1}. The displacement current is

(1) 1 A

(2) 2 A

(3) 3 A

(4) 4 A

7. The focal length of the double convex lens with refractive index 2 and radius of curvature of two surfaces respectively 30 cm and 45 cm is

(1) 16 cm

(2) 18 cm

(3) 30 cm

(4) 40 cm

8. A concave mirror of focal length *f* (in air) is immersed in water ($\mu = \frac{4}{3}$). The focal length of the mirror in water will be

(1) *f*

(2) $\frac{4f}{3}$

(3) $\frac{3f}{4}$

(4) $\frac{7f}{3}$

9. If focal length of a converging lens is 50 cm, then its power is

(1) 5 D

(2) 6 D

(3) 4 D

(4) 2 D

10. An unpolarised light of intensity I_0 is incident on a system of polariser and a analyser. If angle between polariser and analyser is 60°, then intensity of light that comes out from the analyser will be

(1) $I_0/2$

(2) $I_0/4$

(3) $3\ I_0/8$

(4) $I_0/8$

11. Polaroid glass is used in sun glasses because

(1) It has good colour

(2) It is easy to clean

(3) It reduces light intensity to half on account of polarisation

(4) It is having high refractive index

12. The refractive index of medium having critical angle 45° will be

(1) $\sqrt{3}$

(2) 2

(3) $\frac{2}{\sqrt{3}}$

(4) $\sqrt{2}$

13. There are three metal plates *A*, *B* and *C* and their work function are related as $\phi_A > \phi_B > \phi_C$. If the graph is drawn between the maximum kinetic energy of electron vs frequency of incident radiation and slope of the three curves for *A*, *B* and *C* is found to be m_A, m_B and m_C respectively, then

(1) $m_A > m_B > m_C$

(2) $m_A < m_B < m_C$

(3) $m_A = m_B = m_C$

(4) $m_A = 2m_B = m_C$

CC-108 CC-108 CC-108 CC-108 CC-108 CC-108

14. In single slit diffraction, wavelength used is 5000 Å, slit width is 0.2 mm and distance between slit and screen is 2 m. The distance of first secondary maxima from principal maximum is

(1) 2.5 mm

(2) 3 mm

(3) 4.5 mm

(4) 7.5 mm

15. A : In photoelectric effect, on increasing the intensity of light, kinetic energy of fastest electron increases.
R : Kinetic energy of fastest electron is directly proportional to intensity.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

16. A particle of mass M at rest decays into two particles of masses m_1 and m_2 having non-zero velocities. The ratio of the de-Broglie wavelengths of the particles $\lambda_1 : \lambda_2$ is

(1) $m_1 : m_2$

(2) $m_2 : m_1$

(3) 1 : 1

(4) $\sqrt{m_2} : \sqrt{m_1}$

17. Bohr's model is applicable for

(1) He^+

(2) Li^{3+}

(3) Li^+

(4) He

18. The total energy of an electron in its nth excited state of hydrogen atom is K. Its kinetic energy in this state is

(1) $-K$

(2) $K/2$

(3) $2K$

(4) $\frac{2K}{3}$

19. Match List-I with List-II

	List-I		List-II
a.	Frequency of incident radiation increases	(i)	Stopping potential decreases
b.	Number of incident photons increases	(ii)	Stopping potential increases
c.	Intensity of incident radiation decreases	(iii)	Saturation current increases
d.	Maximum K.E. of photoelectrons decreases	(iv)	Number of photoelectrons emitted per second decreases

Choose the correct answer from the options given below:

(1) a(iii), b(ii), c(i), d(iv)

(2) a(i), b(iv), c(iii), d(ii)

(3) a(ii), b(iii), c(iv), d(i)

(4) a(iv), b(i), c(ii), d(iii)

20. If the radius of the innermost electron orbit of a hydrogen orbit is 5×10^{-11} m, then radius of the 5th orbit is

(1) 2×10^{-12} m

(2) 1.25×10^{-10} m

(3) 2.5×10^{-10} m

(4) 1.25×10^{-9} m

21. In the α-particle scattering experiment, if impact parameter b is very large, the angle of deviation θ of α-particle is

(1) 0°

(2) $\frac{\pi}{2}$

(3) $\frac{2\pi}{3}$

(4) π

22. Light from a point source in air falls on a spherical glass surface (μ = 1.5 and radius of curvature = 40 cm). If image is formed at infinity, then distance of the light source from the glass surface is

(1) 20 cm

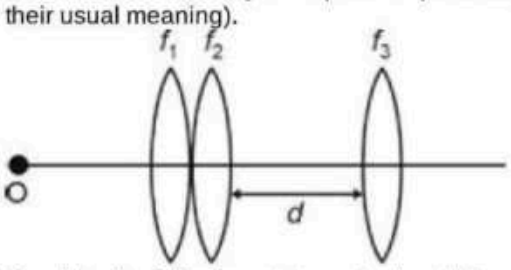
(2) 40 cm

(3) 80 cm

(4) Infinity

CC-108 CC-108 CC-108 CC-108 CC-108 CC-108 CC-108

23. Consider three thin lenses *A*, *B* and *C* which are placed as shown below with object at point *O* (where symbols have their usual meaning).



Consider the following statements about this combination.

(A) Focal length of combination is given by $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3}$

(B) Power of combination, $P = P_1 + P_2 + P_3$

(C) Total magnification of combination, $m = m_1 m_2 m_3$

Correct statements among the above are

(1) (A) and (B) only

(2) (B) and (C) only

(3) (C) only

(4) (A), (B) and (C)

24. In a Young's double slit experiment in front of one of the slits a thin transparent sheet of thickness *t* and refractive index μ is inserted. The entire fringe pattern will be shifted by a distance (*d* – separation between slits; *D* – distance of screen from the slits and λ – wavelength of light used)

(1) $\frac{(\mu-1)tD}{\lambda}$

(2) $\frac{(\mu-1)\lambda D}{d}$

(3) $\frac{(\mu-1)tD}{d}$

(4) $\frac{(\mu-1)td}{D}$

25. Consider the following statements (A) and (B) and identify the correct answer:

Statements (A) : When monochromatic light is incident on a surface separating two media, the reflected and refracted light both have the same frequency as the incident frequency.

Statements (B) : When light travels from a denser to rarer medium, the speed increases.

(1) (A) is correct but (B) is incorrect

(2) (A) is incorrect but (B) is correct

(3) Both (A) and (B) are correct

(4) Both (A) and (B) are incorrect

26. Identify the incorrect maxwell's equation in vacuum.

(1) Gauss's law for magnetism is $\oint \vec{B} . \vec{dA} \neq 0$

(2) Faraday's law is $\oint \vec{E} . d\vec{l} = \frac{-d\phi_B}{dt}$

(3) Ampere–Maxwell law is $\oint \vec{B} . \vec{dl} = \mu_0 i_c + \mu_0 \varepsilon_0 \frac{d\phi_E}{dt}$

(4) Both (2) and (3)

27. The electric field in a plane electromagnetic wave is given by $E_y = (4\times10^7)$ N/C $\sin(100\pi x + 3\pi\times10^{10} t)$ where *x* is in m and *t* is in s. Wavelength of this wave is

(1) 1 cm

(2) 2 cm

(3) 3.14 cm

(4) 3.14 m

28. A container is filled with water upto a height *H*. The apparent depth of a coin lying at the bottom of the container is found to be 9 cm. If refractive index of water is $\frac{4}{3}$, then *H* is

(1) 6 cm

(2) 6.75 cm

(3) 12 cm

(4) 13.5 cm

29. When light moves through an optical fibre, it undergoes

(1) Interference

(2) Diffraction

(3) Total internal reflection

(4) Polarisation

30. The magnifying power of a simple microscope when the final image is formed at *D* = 25 cm from eye is 2. Then its focal length *f* is

(1) +12.5 cm

(2) –12.5 cm

(3) +25 cm

(4) –25 cm

31. A light beam of wavelength 6400 Å is incident on two slits of same size which are at *d* = 0.8 mm apart. Fringes are obtained on a screen at a distance `*D*' from the double slits. If the fifth interference minimum is observed 3.60 mm from central maximum, then fringe width is equal to

(1) 0.8 mm

(2) 0.4 mm

(3) 1 mm

(4) 1.8 mm

32. The minimum energy needed by an electron to come out from a metal surface is called the metal's

(1) Stopping potential

(2) Work function

(3) Both (1) and (2)

(4) Neither (1) nor (2)

CC-108 CC-108 CC-108 CC-108 CC-108 CC-108 CC-108
Aakash Medical | IIT-JEE | Foundations

33. The correct method of detection for different types of electromagnetic waves are:
(A) Receiver's aerials for microwaves
(B) Our eyes for visible light
(C) Photocells for infra-red waves
(D) Ionisation chamber for *X*-rays
Choose the correct answer from the options given below:

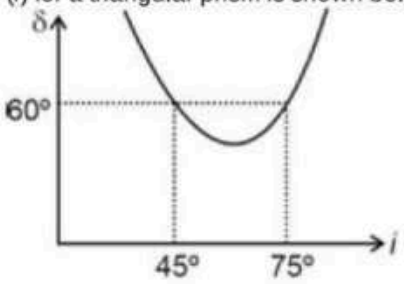
(1) (A) and (C) only

(2) (B) and (D) only

(3) (A), (B) and (D) only

(4) (A), (B), (C) and (D)

34. If the plot of angle of deviation (δ) versus angle of incidence (*i*) for a triangular prism is shown below, then prism angle is



(1) 45°

(2) 60°

(3) 75°

(4) 90°

35. A parallel plate capacitor made of circular plates each of radius $R = 5$ cm has a capacitance $C = 50$ pF. The capacitor is connected to a 220 V AC supply with an angular frequency of 50 rad s^{-1}. Then the *RMS* value of displacement current across the plates is (in μA)

(1) 0.275

(2) 0.55

(3) 1.1

(4) 2.2

CC-108 CC-108 CC-108 CC-108 CC-108 CC-108 CC-108

SECTION-B

36. Instantaneous value of electric current varies in a wire according to $i = [8 + 15\sin(200\ \pi t)]$ ampere. Its average value over one complete cycle is given as

(1) 6 A

(2) 8 A

(3) 23 A

(4) 2.3 A

37. An electromagnetic pulse having intensity of 1200 W/m^2 is incident on a metal surface. If pulse after normal incidence over the plate reflects back completely, then force applied by pulse on the metal surface will be (Given surface area of metal is 2 m^2)

(1) 0.8×10^{-5} N

(2) 0.8×10^{5} N

(3) 1.6×10^{5} N

(4) 1.6×10^{-5} N

38. The oscillating magnetic field in an electromagnetic wave is given by $B_y = 4 \times 10^{-6} \sin(2000\pi x - 6 \times 10^{11}\pi t)$ T. The frequency of the electromagnetic wave is

(1) 6×10^{11} Hz

(2) 3×10^{11} Hz

(3) $6\pi \times 10^{11}$ Hz

(4) $3\pi \times 10^{11}$ Hz

39. A compound microscope has a magnifying power 30. Focal length of its eye piece is 5 cm. If the final image formed at least distance of distinct vision then magnification produced by the objective is

(1) 2

(2) 6

(3) 5

(4) 3

40. The coherence of two light sources means that the light waves emitted have

(1) Same wavelength

(2) Same intensity

(3) Constant phase difference

(4) Same velocity

41. Work function of a metal is 2.2 eV. A photon of energy 3.8 eV is incident on it. The maximum kinetic energy of emitted electron will be

(1) 1 eV

(2) 2 eV

(3) 1.6 eV

(4) 2.6 eV

42. Which of the following transition for He^+ atom will have shortest wavelength?

(1) $n = 3$ to $n = 2$

(2) $n = 15$ to $n = 4$

(3) $n = 2$ to $n = 1$

(4) $n = 15$ to $n = 2$

43. In which of the following physical processes, electron emission occurs from metal surface?

(1) Thermionic emission

(2) Field emission

(3) Photoelectric emission

(4) All of these

44. Let I_0 be the intensity of polarised light after passing through the first polariser P_1 and it also passes through polaroids P_2 and P_3. If θ is the angle between pass axes of P_1 and P_2 while $\left(\frac{\pi}{2}-\theta\right)$ is angle between pass axes of P_2 and P_3, then transmitted intensity of emergent light from P_3 will be

(1) $\frac{I_0}{4}\sin^2(2\theta)$

(2) $\frac{I_0}{4}\cos^2(2\theta)$

(3) $\frac{I_0}{2}\sin^2(2\theta)$

(4) $\frac{I_0}{2}\cos^2(2\theta)$

45. Monochromatic light of frequency 1.5×10^{18} Hz is produced by a source. If 2×10^{15} photons per second (on an average) are emitted by the source, then power of the source is nearly

(1) 2 mW

(2) 2 W

(3) 2 kW

(4) 2 MW

46. The average power consumed by an inductor over one complete cycle in an AC circuit is [symbols have their usual meaning]

(1) $\frac{i_m V_m}{2}$

(2) $i_m V_m$

(3) Zero

(4) $\frac{i_m V_m}{4}$

47. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R.**

Assertion (A) : In a purely inductive or capacitive circuit, $\cos\phi = 1$

Reason (R) : In a purely inductive or capacitive circuit no power is dissipated even though a current is flowing in the circuit.

In the light of the above statements, the correct option is

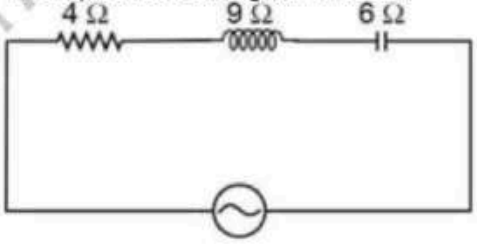
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

48. The impedance of the given circuit is



(1) 5 Ω

(2) 9 Ω

(3) 6 Ω

(4) Zero

CC-108 CC-108 CC-108 CC-108 CC-108 CC-108
Medical | IIT-JEE | Foundation

49. The inductive reactance is

(1) Directly proportional to inductance

(2) Inversely proportional to inductance

(3) Directly proportional to frequency of current

(4) Both (1) and (3)

50. The instantaneous value of current for $I = I_0 \sin\omega t$ at $t = \frac{T}{8}$ is

(1) $\sqrt{2}I_0$

(2) $\sqrt{3}I_0$

(3) $2I_0$

(4) $\frac{I_0}{\sqrt{2}}$

CHEMISTRY

SECTION-A

CC-108 CC-108 CC-108 CC-108 CC-108 CC-108 CC-108

51. If, on increasing the temperature from 310 K to 320 K, the rate of the reaction gets doubled, then the activation energy of the reaction is

(1) 9527 R ln2

(2) 8250 R ln2

(3) 10982 R ln2

(4) 9920 R ln2

52. 50% of a first order reaction was completed in 6.93 min, then 75% of the same reaction would be completed in

(1) 2.303 min

(2) 13.86 min

(3) 69.09 min

(4) 23.03 min

53. The rate of a first order reaction is 2.5×10^{-3} mol L^{-1} min^{-1} at 0.25 M concentration of reactant. The half life (in min) of the reaction is

(1) 69.3

(2) 43.5

(3) 76.6

(4) 38.7

54. If rate constant 'k' of a reaction varies with temperature as $\ln k = 2 - \frac{200}{T}$, then Arrhenius constant (A) and activation energy (E_a) of the reaction respectively are

(1) 100, 200 R

(2) 100, 20 R

(3) e^2, 200 R

(4) e^2, 20 R

55. If rate of disappearance of H_2O_2 during its decomposition is 0.2 mol $L^{-1}s^{-1}$ then rate of appearance of O_2 is (given : $H_2O_2 \rightarrow H_2O + 1/2\ O_2$)

(1) 0.2 mol $L^{-1}s^{-1}$

(2) 0.1 mol $L^{-1}s^{-1}$

(3) 0.3 mol $L^{-1}s^{-1}$

(4) 0.4 mol $L^{-1}s^{-1}$

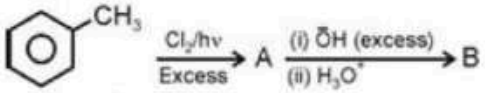
56. Which one is most reactive towards S_N1 reaction?

(1) $C_6H_5CH_2Br$

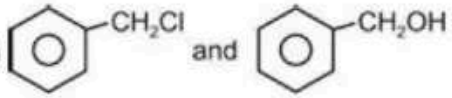
(2) $(C_6H_5)_2CHBr$

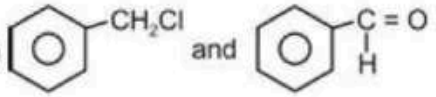
(3) $(C_6H_5)_3CBr$

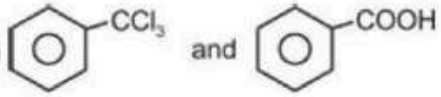
(4) C_6H_5Br

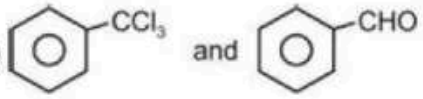
57. 

Products A and B respectively are

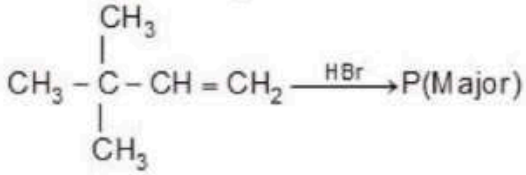
(1) 

(2) 

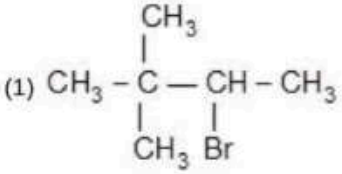
(3) 

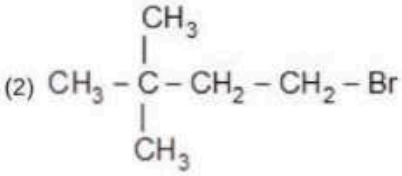
(4) 

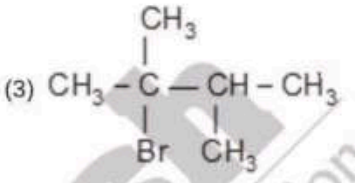
58. Consider the following reaction

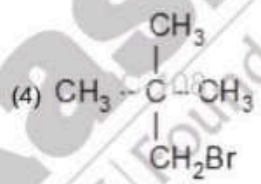


Major product P is

(1) 

(2) 

(3) 

(4) 

59. Among the following which methyl halide is of maximum dipole moment?

(1) CH_3F

(2) CH_3Cl

(3) CH_3Br

(4) CH_3I

60. In the following reaction, $C_6H_5Br \xrightarrow[(ii)C_2H_5OH]{(i)Mg,ether} A$, the product (A) is

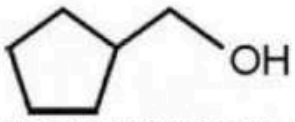
(1) C_6H_6

(2) $C_6H_5 - C_2H_5$

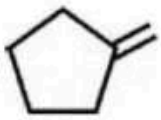
(3) $C_6H_5 - OH$

(4) $C_6H_5OC_2H_5$

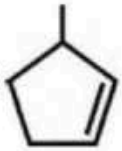
CC-108 CC-108 CC-108 CC-108 CC-108 CC-108 CC-108

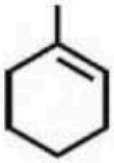
61. 

on dehydration with conc. H_2SO_4 predominantly forms

(1) 

(2) 

(3) 

(4) 

62. Which of the following reaction mixtures does not produce 1° alcohol?

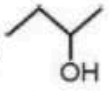
(1) HCHO + RMgX; followed by hydrolysis

(2) R – CHO + H_2; over palladium

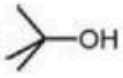
(3) R – CHO + RMgX; followed by hydrolysis

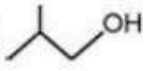
(4) $CH_3 - CH = CH_2 + B_2H_6$; followed by oxidation in the presence of $H_2O_2/OH^{\ominus}$

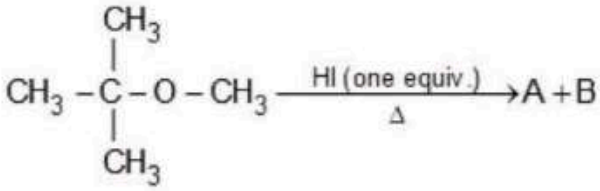
63. The alcohol which reacts fastest by conc. HCl and $ZnCl_2$ is

(1) 

(2) 

(3) 

(4) 

64. 

Major products A and B are

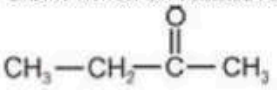
(1) $(CH_3)_3C$ – OH and CH_3OH

(2) $(CH_3)_3C$ – I and CH_3I

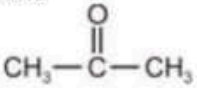
(3) $(CH_3)_3C$ – I and CH_3OH

(4) $(CH_3)_3C$ – OH and CH_3I

65. Olefin which on reductive ozonolysis gives



and



is

(1) 2, 3-Dimethylpent-1-ene

(2) 1, 3-Dimethylpent-1-ene

(3) 2, 3-Dimethylpent-2-ene

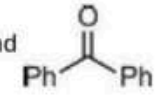
(4) 2, 4-Dimethylpent-2-ene

66. The pair of compound which can be distinguished by I_2/NaOH is

(1) HCHO and CH_3OH

(2) CH_3CHO and C_2H_5OH

(3) C_2H_5OH and PhCHO

(4) CH_3OH and 

67. Given below are two statements

Statement I: Ethanal is miscible with water in all proportions.

Statement II: Ethanal forms hydrogen bonding with water.

In light of above statements, choose the correct answer from the options given below

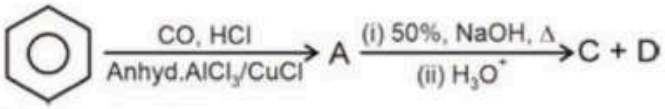
(1) Statement I is incorrect but statement II is correct

(2) Both statement I and statement II are incorrect

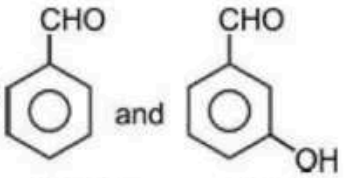
(3) Both statement I and statement II are correct

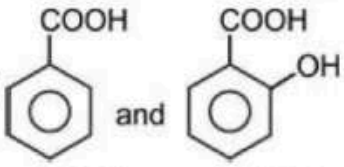
(4) Statement I is correct but statement II is incorrect

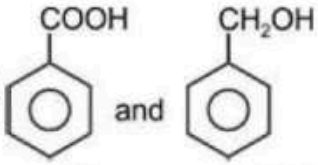
CC-108 CC-108 CC-108 CC-108 CC-108 CC-108 CC-108
Aakash Medical|IIT-JEE|Foundations

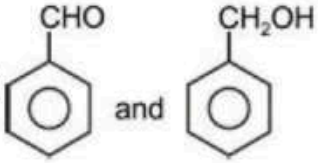
68. 

Products C and D are

(1) 

(2) 

(3) 

(4) 

69. Match List-I with List-II

	List-I Reactions		List-II Order of reaction
a.	$CHCl_3 + Cl_2 \rightarrow CCl_4 + HCl$	(i)	Zero
b.	$2NH_3(g) \xrightarrow[\text{Pt catalyst}]{1130K} N_2(g) + 3H_2(g)$	(ii)	1
c.	$C_2H_4(g) + H_2(g) \rightarrow C_2H_6(g)$	(iii)	Pseudo first
d.	$C_{12}H_{22}O_{11} + H_2O \xrightarrow{H^+} \underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Fructose}}{C_6H_{12}O_6}$	(iv)	1.5

The correct match is

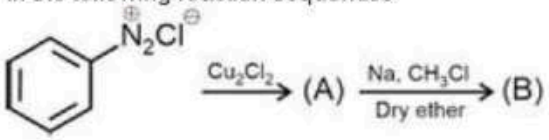
(1) a(iv), b(i), c(ii), d(iii)

(2) a(ii), b(i), c(iii), d(iv)

(3) a(i), b(ii), c(iii), d(iv)

(4) a(iii), b(ii), c(i), d(iv)

70. In the following reaction sequences



The major product (B) and name of reaction from A $\rightarrow$ B formation respectively will be

(1) Toluene and Wurtz-Fittig reaction

(2) Chlorobenzene and Wurtz-Fittig reaction

(3) Toluene and Wurtz reaction

(4) Chlorobenzene and Wurtz reaction

71. Given below are two statements

Statement-I : Reaction of tert-butyl bromide with aqueous KOH yields tert-butyl alcohol as a major product.

Statement-II : Reaction of tert-butyl bromide with aqueous KOH follows the S_N2 mechanism

In the light of above statements, choose the most appropriate answer from the options given below

(1) Both statement-I and statement-II are correct

(2) Both statement-I and statement-II are incorrect

(3) Statement-I is correct but statement-II is incorrect

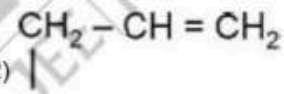
(4) Statement-I is incorrect but statement-II is correct

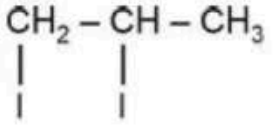
72. In the following reaction

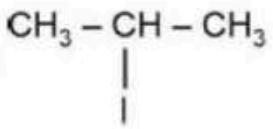
$$CH_3-CH=CH_2 + HCl \rightarrow \underset{\text{(Major)}}{(A)} \xrightarrow[\text{dry acetone}]{\text{NaI in}} \underset{\text{(Major)}}{(B)}$$

The product (B) will be

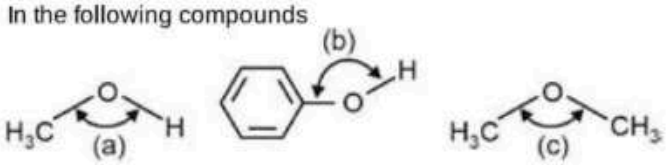
(1) $CH_3 - CH_2 - CH_2 - I$

(2) 

(3) 

(4) 

73. In the following compounds



The correct order of bond angle will be

(1) a > c > b

(2) c > b > a

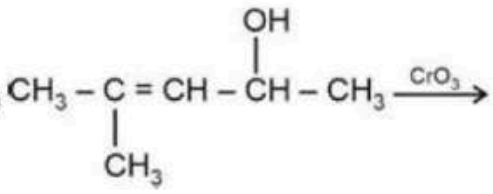
(3) a > b > c

(4) b > c > a

CC-108 CC-108 CC-108 CC-108 CC-108 CC-108 CC-108
Aakash Medical|IIT-JEE|Foundations

74. Which of the following reaction gives crotonaldehyde as a major product?

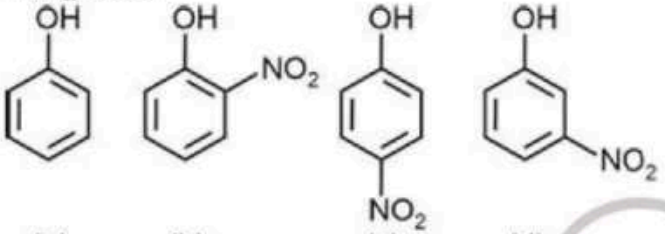
(1) $CH_3 - CH = CH - CH_2 - CH_2 - CN \xrightarrow[(ii)\ H_2O]{(i)\ A/H\ (i-Bu)_2}$

(2) 

(3) $CH_2 = CH - CH_2 - OH \xrightarrow{CrO_3}$

(4) $CH_3 - CH = CH - CH_2 - OH \xrightarrow{PCC}$

75. Arrange the following compounds in their decreasing order of pK_a values



(a) (b) (c) (d)

(1) (a) > (d) > (b) > (c)

(2) (b) > (c) > (d) > (a)

(3) (a) > (b) > (d) > (c)

(4) (b) > (d) > (c) > (a)

76. Given below are two statements:

Statement-I : Reaction of ethyl chlorides with sodium ethoxide majorly gives diethyl ether.

Statement-II : Williamson synthesis reaction involves S_N1 attack of an alkoxide ion on primary alkyl halide

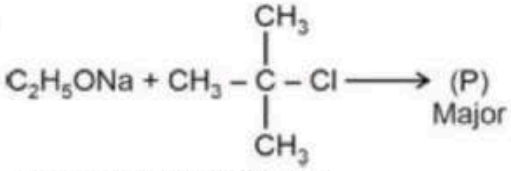
In the light of above statements, choose the most appropriate answer from the options given below

(1) Both statement-I and statement-II are correct

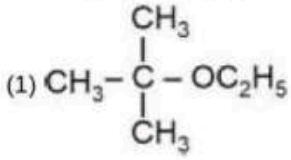
(2) Both statement-I and statement-II are incorrect

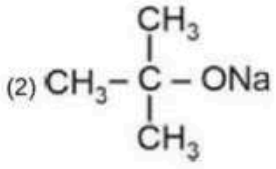
(3) Statement-I is correct but statement-II is incorrect

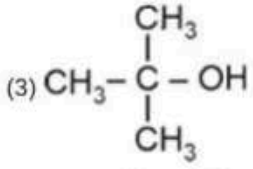
(4) Statement-I is incorrect but statement-II is correct

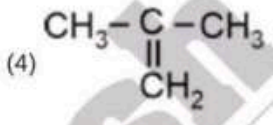
77. 

The major product (P) will be

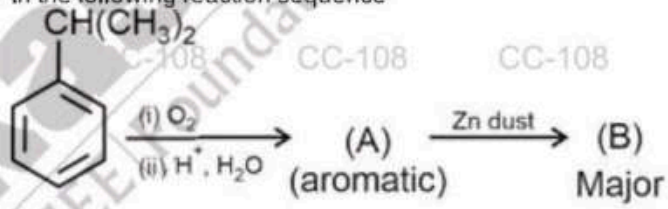
(1) 

(2) 

(3) 

(4) 

78. In the following reaction sequence



The compound (B) will be

(1) Benzene

(2) Phenol

(3) Toluene

(4) Benzoquinone

79. Given below are two statements

Statement-I : Acetaldehyde is more reactive than acetone towards nucleophilic addition reaction.

Statement-II : Benzaldehyde is less reactive than propanal towards nucleophilic addition reaction.

In the light of above statements, choose the most appropriate answer from the options given below

(1) Both statement-I and statement-II are correct

(2) Both statement-I and statement-II are incorrect

(3) Statement-I is correct but statement-II is incorrect

(4) Statement-I is incorrect but statement-II is correct

CC-108 CC-108 CC-108 CC-108 CC-108 CC-108 CC-108

80. In the following reaction sequences

$$CH_3OH \xrightarrow{SOCl_2} (A) \xrightarrow[\substack{(ii)\ H^+ \text{ or } OH^- \\ H_2O, \Delta}]{(i)\ KCN} (B) \xrightarrow{P_2O_5, \Delta} (C)$$

The major product (C) will be

(1) $CH_3 - COOH$

(2) $CH_3 - CH_2 - Cl$

(3) $CH_3 - CH_2 - OH$

(4) $CH_3 - \overset{O}{\overset{\|}{C}} - O - \overset{O}{\overset{\|}{C}} - CH_3$

81. In the following reaction sequences

$$CH_3 - C \equiv CH + H_2O \xrightarrow[333K]{Hg^{2+}/H^+} (A)$$

$$(A) \xrightarrow[(ii)\ KOH/\text{ethylene glycol, heat (Major)}]{(i)\ NH_2-NH_2} (B)$$

The compound (B) and name of reaction from A → B respectively are

(1) $CH_3 - CH_2 - CH_3$ and Wolff-Kishner reduction

(2) $CH_3 - \overset{O}{\overset{\|}{C}} - CH_3$ and Wolff Kishner reduction

(3) $CH_3 - CH_2 - CH_3$ and clemmensen reduction

(4) $CH_3 - \underset{CH_3}{\overset{O}{\overset{\|}{C}}}$ and clemmensen reduction

82. Given below are two statements one is labelled as Assertion (A) and other is labelled as Reason (R)
Assertion (A) : Boiling points of acetone is higher than the propan-1-ol
Reason (R) : Acetone has stronger intermolecular hydrogen bonding than the propan-1-ol.
In the light of above statements, choose the most appropriate answer from the options given below.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is the correct explanation of (A)

(3) Both (A) and (R) are false

(4) (A) is true but (R) is false

83. In the following reaction sequence

$$CH_3 - CH_2 - OH \xrightarrow[\substack{(ii)\ Cl_2/\text{Red phosphorus} \\ (iii)\ H_2O}]{(i)\ CrO_3 - H_2SO_4} \underset{\text{Major}}{(A)}$$

The compound (A) is

(1) $CH_3 - CHO$

(2) $CH_3 - \overset{O}{\overset{\|}{C}} - Cl$

(3) $\underset{Cl}{\overset{CH_2 - COOH}{|}}$

(4) $\underset{Cl}{\overset{CH_2 - \overset{O}{\overset{\|}{C}} - Cl}{|}}$

84. An organic compound (A) does not reduce Tollen's reagent but forms an addition compound with sodium hydrogen sulphite and gives positive iodoform test. On vigorous oxidation it gives ethanoic and propanoic acid. The possible structure of the compound (A) will be

(1) $CH_3 - \overset{O}{\overset{\|}{C}} - CH_2 - CH_2 - CH_3$

(2) $CH_3 - \underset{O}{\overset{CH_3}{\overset{|}{C}}} - CH - CH_3$

(3) $CH_3 - CH_2 - \underset{O}{\overset{}{\overset{\|}{C}}} - CH_2 - CH_3$

(4) $CH_3 - CH_2 - CH_2 - CH_2 - CHO$

85. Given below are two statements:
Statement-I : Carboxylic acids are higher boiling liquids than aldehydes.
Statement-II : Most carboxylic acids exist as dimer in the vapour phase or in the aprotic solvents.
In the light of above statements, choose the most appropriate answer from the options given below

(1) Both statement-I and statement-II are correct

(2) Both statement-I and statement-II are incorrect

(3) Statement-I is correct but statement-II is incorrect

(4) Statement-I is incorrect but statement-II is correct

CC-108 CC-108 CC-108 CC-108 CC-108 CC-108 CC-108

SECTION-B

86. A zero order reaction has initial concentration of reactant 0.1 M and rate constant 0.002 mol L^{-1} s^{-1}. Left concentration of reactant after 30 seconds is

(1) 0.02 M

(2) 0.08 M

(3) 0.04 M

(4) 0.03 M

87. A substance decays by first order kinetics. If the half life of the substance is 16 min then what percentage of the substance remains unchanged after 64 min?

(1) 16.25%

(2) 6.25%

(3) 12.25%

(4) 93.75%

88. Mechanism of a hypothetical reaction $2A_3 \rightarrow 3A_2$ is given below

(i) $A_3 \rightleftharpoons A_2 + A$ (fast)

(ii) $A_3 + A \rightarrow 2A_2$ (slow)

The overall order of the reaction will be

(1) 0

(2) 3

(3) 1

(4) 2

89. The rate of a gaseous reaction is given by the expression $r = k[A][B]^2$. If concentrations of A and B increases by factor of 2 then rate of reaction becomes

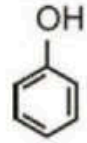
(1) 2 times

(2) 4 times

(3) 8 times

(4) 16 times

90. Arrange the following molecules in decreasing order of their K_a values

(I) 

(II) CH_3COOH

(III) HCOOH

(IV) CH_3OH

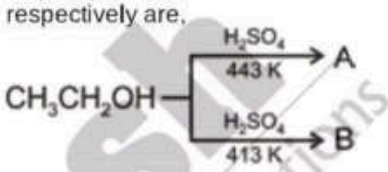
(1) I > II > III > IV

(2) III > II > I > IV

(3) II > I > III > IV

(4) I > III > II > IV

91. Major products A and B in the following reactions respectively are.



(1) $CH_2 = CH_2$ and CH_3CH_3

(2) $CH_2 = CH_2$ and $C_2H_5OC_2H_5$

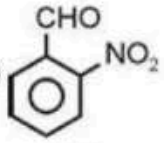
(3) $C_2H_5OC_2H_5$ and $CH_2 = CH_2$

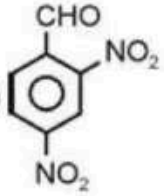
(4) $CH_2 = CH_2$ and CH_3OCH_3

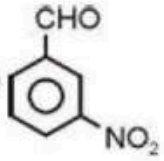
CC-108 CC-108 CC-108 CC-108 CC-108 CC-108 CC-108

92. Benzaldehyde on reaction with HNO_3/H_2SO_4 at 273 – 283 K majorly gives

(1) 

(2) 

(3) 

(4) 

93. Given below are two statements one is labelled as Assertion (A) and other is labelled as Reason (R)

Assertion (A) : Boiling points of methyl halides decreases in the order : $CH_3I > CH_3Br > CH_3Cl > CH_3F$

Reason (R) : With the increase in size and mass of halogen atom, the magnitude of vander Waal forces increases.

In the light of above statements, choose the most appropriate answer from the options given below.

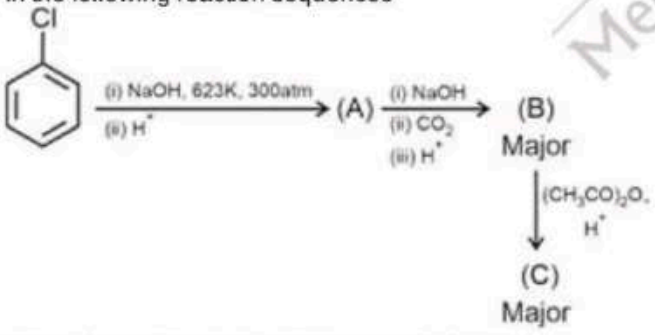
(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is the correct explanation of (A)

(3) Both (A) and (R) are false

(4) (A) is true but (R) is false

94. In the following reaction sequences



The major product (B) and (C) respectively are

(1) p-hydroxybenzoic acid and Aspirin

(2) m-hydroxybenzoic acid and Acetophenone

(3) o-hydroxybenzoic acid and Aspirin

(4) p-hydroxybenzaldehyde and Acetophenone

95. Reaction of propanone with methyl magnesium bromide followed by hydrolysis gives compound (P). Compound (P) on treating with 20% H_3PO_4 at 358 K gives compound (Q). Compound (Q) with Bayer's reagent gives compound (R). The compound (R) will be

(1) 2-Methyl propane-1, 2-diol

(2) 2-Methyl propan-1-ol

(3) Butan-2-one

(4) 2-Methyl propane-1, 3 diol

96. Given below are two statements:

Statement-I : At 273K, reaction of phenol with bromine in CS_2 majorly gives p-bromophenol.

Statement-II : Phenol is treated with bromine water majorly gives 2, 4, 6-tribromophenol

In the light of above statements, choose the most appropriate answer from the options given below

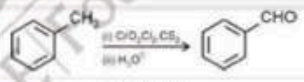
(1) Both statement-I and statement-II are correct

(2) Both statement-I and statement-II are incorrect

(3) Statement-I is correct but statement-II is incorrect

(4) Statement-I is incorrect but statement-II is correct

97. Match List-I with List-II

	List-I Reactions		List-II Name of reaction
a.	$CH_3 - Br + AgF \rightarrow CH_3 - F$	(i)	Fittig
b.		(ii)	Stephen
c.		(iii)	Etard
d.	$CH_3 - CN \xrightarrow[(ii)\ H_3O^+]{(i)\ SnCl_2+HCl} CH_3 - CHO$	(iv)	Swarts

Choose the correct match

(1) a(ii), b(iii), c(i), d(iv)

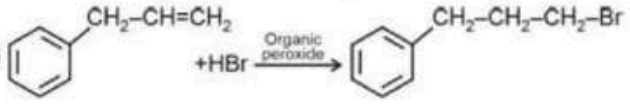
(2) a(iv), b(i), c(iii), d(ii)

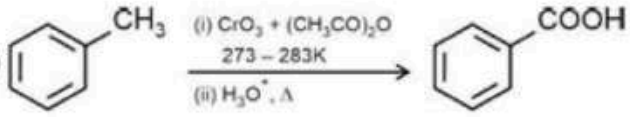
(3) a(i), b(ii), c(iii), d(iv)

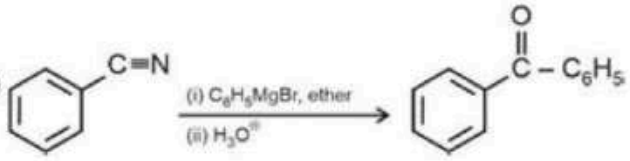
(4) a(iii), b(iv), c(ii), d(i)

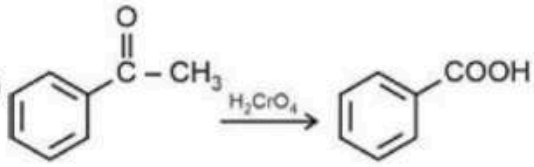
Aakash Medical | IIT-JEE | Foundations

98. Select the incorrect reaction among the following:

(1) 

(2) 

(3) 

(4) 

99. Given below are two statements

Statement-I : Benzaldehyde and acetaldehyde can be distinguished by Fehling's test.

Statement-II : Both benzaldehyde and acetaldehyde give Cannizzaro reaction

In the light of above statements, choose the most appropriate answer from the options given below

(1) Both statement-I and statement-II are correct

(2) Both statement-I and statement-II are incorrect

(3) Statement-I is correct but statement-II is incorrect

(4) Statement-I is incorrect but statement-II is correct

100. In the following reaction sequences

$$CH_3-CH_2-MgCl \xrightarrow[(ii)\ H_3O^+]{(i)\ CO_2,\ dry\ ether} (A) \xrightarrow[(ii)\ H_2,\ Pd-BaSO_4\ Major]{(i)\ SOCl_2} (B)$$
$$(iii)\ dil.\ NaOH$$
$$(iv)\ \Delta$$

The compound (B) will be

(1) 3-hydroxy-4-Methyl pentanal

(2) 2-Methyl pent-3-enal

(3) 2-Methyl pent-2-enal

(4) Propanal

CC-108 CC-108 CC-108 CC-108 CC-108 CC-108 CC-108

BOTANY

SECTION-A

101. Which of the following traits of pea plant will be expressed only in the homozygous form?

(1) Violet flower colour

(2) Tall stem height

(3) Inflated pod shape

(4) Terminal flower position

102. Among the given options, an example of test cross is

(1) RrYy × RRYY

(2) Tt × Tt

(3) RrYy × rryy

(4) Tt × TT

103. The below given scientists independently rediscovered Mendel's work on inheritance of characters, **except**

(1) de Vries

(2) Correns

(3) R.C. Punnett

(4) von Tschermak

104. Which of the following will **not** be observed in a dinucleotide?

(1) N-glycosidic linkage

(2) 3'-5' phosphodiester linkage

(3) Hydrogen bonds

(4) Phosphoester linkage

105. In the double helix structure of DNA, the pitch of the helix is _(A)_ and there are roughly _(B)_ bp in each turn.
Select the **correct** option for (A) and (B) respectively.

	(A)	(B)
(1)	3.4 A°	11 bp
(2)	0.034 nm	9 bp
(3)	0.34 nm	10 bp
(4)	3.4 nm	10 bp

(1) (1)

(2) (2)

(3) (3)

(4) (4)

106. Which of the following is **incorrect** w.r.t double helical structure of DNA?

(1) The two chains of DNA have anti-parallel polarity

(2) Apart from the hydrogen bonds, the stacking of one base pair over the other in the double helix provides stability to the helical structure

(3) It is made up of two polynucleotide chains, where the backbone is constituted by sugar-phosphate and the bases project outside.

(4) The bases in two strands are paired through hydrogen bonding.

107. Single stranded DNA is the genetic material in

(1) Bacteriophage λ

(2) *E. coli*

(3) Bacteriophage $\phi \times 174$

(4) *Drosophila melanogaster*

108. Match **List I** with **List II**

	List I		List II
a.	Incomplete dominance	(i)	Phenylketonuria
b.	Codominance	(ii)	Myotonic dystrophy
c.	Autosomal dominant trait	(iii)	Phenotype that did not resemble either of the two parents and was in between the two
d.	Pleiotropy	(iv)	F_1 generation resembles both parents

Select the **correct** answer from the following options.

(1) a(iv), b(i), c(iii), d(ii)

(2) a(iii), b(iv), c(ii), d(i)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(iii), b(ii), c(i), d(iv)

109. Experimental verification of the chromosomal theory of inheritance was done by

(1) Walter Sutton

(2) Theodore Boveri

(3) T.H. Morgan

(4) Alfred Sturtevant

110. Select the **incorrect** statement w.r.t polygenic inheritance.

(1) Such traits are generally controlled by three or more genes

(2) Human skin colour is a classic example of this type of inheritance

(3) Human with genotype, AABBCC will have the darkest skin colour and aabbcc will have the lightest skin colour

(4) Only the dominant allele determines the phenotype of an individual

111. If a man with 'A' blood group marries a woman with 'B' blood group. Which of the following is **not** true?

(1) The 1st child can have an AB or O blood group

(2) The 2nd child can have blood group with I^AI^O or I^BI^O genotype

(3) If the mother has I^BI^B genotype then 'O' blood group is not possible in any of the children.

(4) Their child can have 'A' blood group with I^AI^A genotype.

112. Law of independent assortment **cannot** be observed in a

(1) Monohybrid cross

(2) Dihybrid cross

(3) Trihybrid cross

(4) Tetrahybrid cross

CC-108 CC-108 CC-108 CC-108 CC-108 CC-108
Aakash Medical | IIT-JEE | Foundations

113. Which of the following organism shows the haplodiploid sex-determination?

(1) Bird

(2) *Drosophila*

(3) Grasshopper

(4) Honey bee

114. Select the **incorrect** match

(1)	Discovery of X body	–	Henking
(2)	XO type of sex determination	–	Chicken
(3)	Alfred Sturtevant	–	Mapping of the position of genes on the chromosome
(4)	Mendelian disorder	–	Cystic fibrosis

115. Central Dogma of molecular biology was proposed by

(1) James Watson

(2) Francis Crick

(3) Frederick Griffith

(4) Walter Sutton

116. Read the following statements regarding the "Packaging of DNA"

(a) In prokaryotes, such as *E.coli*, DNA is scattered throughout the cell.

(b) The DNA in nucleoid is organised in large loops held by proteins.

(c) Nucleosomes are formed by wrapping of negatively charged DNA around the histone octamer.

(d) In eukaryotes, chromatin fibres are further coiled and get fully condensed at interphase stage to form chromosomes.

(e) The packaging of chromatin in "beads-on-string" structures requires additional set of proteins called the non-histone chromosomal proteins.

Choose the **correct** ones.

(1) (a), (c) and (e) only

(2) (b), (c) and (d) only

(3) (a), (c) and (d) only

(4) (b) and (c) only

117. Heterochromatin is

(1) Loosely packed region of DNA

(2) Lightly stained region of DNA

(3) Transcriptionally active chromatin

(4) Transcriptionally inactive chromatin

118. Read the following statements and choose the **correct** option.

Statement (A) : Mendel's work suggested that factors were discrete units, he could not provide any physical proof for their existence.

Statement (B) : A cross between red flower and pink flower of snapdragon, results into red and pink flowers in the ratio of 3 : 1 respectively.

(1) Both statement (A) and statement (B) are correct

(2) Both statement (A) and statement (B) are incorrect

(3) Statement (A) is correct but statement (B) is incorrect

(4) Statement (A) is incorrect but statement (B) is correct

119. How many different types of gametes will be produced by an organism having genotype AaBb?

(1) 1

(2) 2

(3) 4

(4) 6

120. Which of the following is **incorrect** w.r.t *Drosophila* as suitable material for genetic experiments?

(1) It could be grown on simple synthetic medium in the laboratory

(2) The fly has short life cycle of about two weeks

(3) A single mating could produce very small number of progeny

(4) There is a clear differentiation of the male and female sex

121. The unequivocal proof that DNA is the genetic material came from the experiments of

(1) Erwin Chargaff

(2) Sutton and Boveri

(3) Frederick Griffith

(4) Alfred Hershey & Martha Chase

122. *E.coli* was grown in the medium containing $^{14}NH_4Cl$ and after growing for many generations, it was transferred into a medium containing N^{15}. What will be ratio of light, hybrid and heavy DNA respectively at the end of third generation of *E.coli*?

(1) 0 : 1 : 4

(2) 0 : 1 : 3

(3) 0 : 1 : 6

(4) 1 : 1 : 4

CC-108 CC-108 CC-108 CC-108 CC-108 CC-108 CC-108

123. The average rate of polymerisation of DNA during the process of DNA replication in *E. coli* is approximately

(1) 1000 bp/sec

(2) 2000 bp/sec

(3) 500 bp/sec

(4) 500 bp/min

124. State **True (T)** or **False (F)** and select the **correct** answer from the following options.

(a) In pleiotropy, a single gene affects multiple metabolic pathways.

(b) In birds, females have a pair of Z chromosomes, whereas male has one Z and one W chromosome.

(c) Frame-shift mutations are caused by deletions and insertions of one or two bases of DNA.

(d) Mendelian disorders are caused due to failure of segregation of chromatids during cell cycle resulting in the gain or loss of a chromosome.

	(a)	(b)	(c)	(d)
(1)	T	F	T	F
(2)	F	T	T	F
(3)	T	F	F	F
(4)	T	F	T	T

(1) (1)

(2) (2)

(3) (3)

(4) (4)

CC-108 CC-108 CC-108 CC-108 CC-108 CC-108 CC-108

125. Chromosome complement of an individual, suffering from Klinefelter's syndrome is

(1) 47, XXX

(2) 45, X0

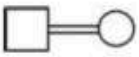
(3) 47, XXY

(4) 46, XY

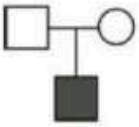
126. (Deleted)

127. Select the **incorrect** match

(1)  – Sex unspecified

(2)  – Consanguineous mating

(3)  – Affected male

(4)  – Male child having affected parents

128. A male having sickle cell anemia marries to a female, who is the carrier of this disease. What percentage of the progeny will have this disorder?

(1) 75%

(2) 50%

(3) 25%

(4) 0%

129. All of the following features make DNA better genetic material, **except.**

(1) DNA is chemically less reactive

(2) DNA is more stable as it has thymine at the place of uracil

(3) It provides scope of rapid changes

(4) It express itself in the form of Mendelian characters

130. Taylor and colleagues proved semiconservative mode of replication in *Vicia faba* using

(1) N^{15} labelled DNA

(2) Radioactive labelled thymidine

(3) P^{32} labelled DNA

(4) S^{35} labelled DNA

131. In a mendelian dihybrid cross of pea plants when F_1 generation is selfed, the probability of getting seeds with genotype Rryy, RrYy and rrYy in F_2 generation respectively will be

(1) $\frac{1}{4} : \frac{1}{8} : \frac{1}{8}$

(2) $\frac{2}{16} : \frac{2}{16} : \frac{2}{16}$

(3) $\frac{4}{16} : \frac{4}{16} : \frac{2}{16}$

(4) $\frac{1}{8} : \frac{1}{4} : \frac{1}{8}$

132. Physical association of genes on the chromosome is called

(1) Sex determination

(2) Recombination

(3) Crossing over

(4) Linkage

133. How many of the following are Mendelian disorders?
Phenylketonuria, Sickle cell anemia, Colour blindness, Down's syndrome, Thalassemia, Turner's syndrome

(1) 3

(2) 4

(3) 2

(4) 1

134. Read the following statements w.r.t the process of replication
(a) Deoxyribonucleoside triphosphates, serve dual purpose of acting as substrates and providing energy for the polymerization reaction.
(b) DNA polymerases can initiate the process of replication on their own.
(c) A failure in cell division (cytokinesis) after DNA replication results into aneuploidy.
(d) DNA polymerases require a DNA template to catalyse the polymerization of ribonucleotides.
(e) Energetically replication is a very expensive process.
Select the **correct** ones.

(1) (a), (c) and (e) only

(2) (b) and (d) only

(3) (a) and (e) only

(4) (b), (c) and (d) only

135. In the experiments, conducted by Griffith, which of the following caused the death of the mice?
(a) Injecting S strain into live mice
(b) Injecting R strain into live mice
(c) Injecting S strain (Heat killed) and live R strain into mice
Select the **correct** ones.

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a) and (c) only

(4) (a), (b) and (c) only

CC-108 CC-108 CC-108 CC-108 CC-108 CC-108 CC-108

SECTION-B

136. During replication of DNA, the discontinuously synthesized fragments are later joined by an enzyme

(1) DNA ligase

(2) DNA polymerase

(3) RNA polymerase

(4) DNase

137. Read the following statements and choose the **correct** option.
Statement (A) : The gene controlling starch grain size in pea shows incomplete dominance.
Statement (B) : Morgan united the knowledge of chromosomal segregation with mendelian principles and called it chromosomal theory of inheritance.

(1) Both statement (A) and statement (B) are correct

(2) Both statement (A) and statement (B) are incorrect

(3) Statement (A) is correct but statement (B) is incorrect

(4) Statement (A) is incorrect but statement (B) is correct

138. In the experiments of Morgan conducted on *Drosophila*, the observed percentage of recombinant types obtained when cross is performed between white eyed miniature winged male and wild type female was

(1) 13%

(2) 37.2%

(3) 50%

(4) 25%

139. Which of the following characteristics of inheritance is shown by blood groups in humans?
(a) Dominance
(b) Polygenic inheritance
(c) Multiple allelism
(d) Co-dominance
Select the **correct** ones.

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a), (c) and (d) only

(4) (c) and (d) only

140. According to the Chargaff's rule, in a double stranded DNA, if number of adenine = 20%, what is the corresponding percentage of guanine?

(1) 30%

(2) 60%

(3) 35%

(4) 40%

141. Read the following statements and select the **correct** ones w.r.t sickle cell anaemia

(a) It is an autosome linked recessive trait.
(b) It is caused due to a point mutation.
(c) The defect is caused due to substitution of valine by glutamic acid.
(d) There is a base substitution from GUG →GAG.
(e) There is a change in the shape of RBC from biconcave disc to sickle like structure.

(1) (a), (c) and (e) only

(2) (b), (c) and (d) only

(3) (a), (b) and (e) only

(4) (b), (c) and (e) only

142. Select the **incorrect** one for Down's syndrome

(1) Autosomal trisomy

(2) There is an additional copy of chromosome number 21

(3) It was first described by Langdon Down

(4) The individual is short statured and have large head

143. β Thalassemia is caused due to

(1) HBA2 gene on chromosome 16

(2) HBA2 gene on chromosome 11

(3) HBB gene on chromosome 11

(4) HBB gene on chromosome 16

144. Histones are rich in

(1) Arginine and histidine

(2) Glycine and alanine

(3) Arginine and lysine

(4) Lysine and alanine

145. Read the following **Assertion (A)** and **Reason (R)**.

Assertion (A) : In bacteria, many times the translation can begin much before the mRNA is fully transcribed.

Reason (R) : In bacteria, mRNA does not require processing to become active and there is no separation of cytosol and nucleus.

In light of the above statements, choose the correct answer

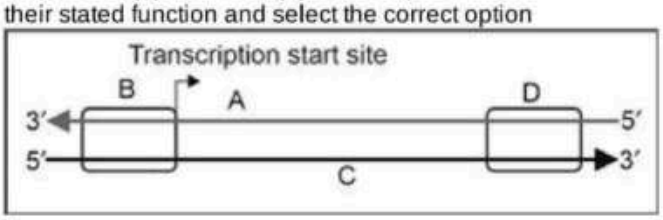
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) Both (A) and (R) are false

(4) (A) is true but (R) is false

146. State **True (T)** or **False (F)** with respect to the labels and their stated function and select the correct option



A – Mostly monocistronic in prokaryotes
B – It is located towards 5' end, with respect to the polarity of the coding strand.
C – In eukaryotes, this strand is used as template for formation of mRNA.
D – It defines the end of the process of transcription.

	A	B	C	D
(1)	T	F	F	T
(2)	F	T	F	T
(3)	T	F	T	F
(4)	T	T	F	T

(1) (1)

(2) (2)

(3) (3)

(4) (4)

147. The contribution of Avery, MacLeod and McCarty is

(1) Discovery of nuclein in the nucleus

(2) Elucidation of DNA packaging process

(3) Discovery of transformation

(4) Biochemical characterization of transforming principle

148. If the sequence of template strand is
5'ATGCGCTATACG 3'
What will be the corresponding mRNA sequence?

(1) 5' AUGCGCUAUACG 3'

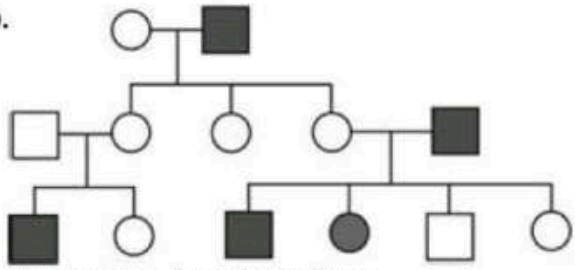
(2) 5' CGUAUAGCGCAU 3'

(3) 5' TACGCGATATGC 3'

(4) 5' CGATATCGCGTA 3'

CC-108 CC-108 CC-108 CC-108 CC-108 CC-108 CC-108
Aakash Medical | IIT-JEE | Foundations

149.



The pedigree given above shows

(1) X-linked dominant inheritance

(2) X-linked recessive inheritance

(3) Autosomal dominant inheritance

(4) Y-linked inheritance

150. A genotypic and phenotypic ratio of 1 : 2 : 1 is observed in case of

(1) Cross between white and violet flowered pea plants

(2) Cross between pea plant with inflated pods and with constricted pods.

(3) Dihybrid test cross

(4) Incomplete dominance

ZOOLOGY

CC-108 CC-108 CC-108 CC-108 CC-108 CC-108 CC-108

SECTION-A

151. An extra-testicular, highly coiled duct which receives openings of vasa efferentia, structurally leads to the

(1) Rete testis

(2) Vas deferens

(3) Epididymis

(4) Seminiferous tubules

152. In humans, one of the cell types present in the intertubular spaces outside the seminiferous tubules are

(1) Sustentacular cells

(2) Immature male germ cells

(3) Spermatogonium

(4) Immunologically competent cells

153. In human females, the widest part of the oviduct, lined by the ciliated columnar epithelium is

(1) Isthmus

(2) Ampulla

(3) Infundibulum

(4) Fimbriae

154. During labor and childbirth, coordinated contractions of the _____ in response to birth hormone released from the _____ pituitary help to expel out the human foetus from the uterus.
Choose the correct option to fill in the blanks respectively.

(1) Endometrium, anterior

(2) Perimetrium, intermediate

(3) Myometrium, anterior

(4) Myometrium, posterior

155. The enlarged end of the human male external genitalia is covered by a loose fold of skin called

(1) Clitoris

(2) Foreskin

(3) Glans penis

(4) Hymen

156. Choose the correct option w.r.t Nirodh.

(1) It's spermicidal in nature

(2) It prevents the entry of ejaculated semen into female reproductive tract

(3) It suppresses the sperm motility in the female genital tract

(4) It releases copper ions

157. Due to encouragement from governmental and non-governmental agencies in reproduction-related areas, which of the following organisations developed a new oral contraceptive for females in India that is chemically non-steroidal?

(1) CDRI

(2) EFB

(3) NACO

(4) WHO

158. Removal of both the ovaries in an adult human female may lead to

(1) Increase in the number of mammary ducts in breasts

(2) Sudden increase in height by several inches

(3) Increased resorption of bones

(4) Increased proliferation of the endometrium

159. Which of the following is the correct statement?

(1) Spermicidal jelly is used along with CuT by females to increase its contraceptive efficiency.

(2) Vault is used with spermicidal foams to cover the penis.

(3) Cervical caps along with spermicidal creams prevent conception by blocking the entry of sperms *via* cervix.

(4) Diaphragm along with foams is used to cover urethral meatus in males.

160. Consider the following statements w.r.t humans:
(a) The mitotic division starts as the zygote moves through the isthmus of the oviduct.
(b) The size of 8-celled morula is larger than the size of a zygote.
(c) Trophoblasts are inner group of embryonic cells that are totipotent in nature.
Which of the above statements is/are incorrect?

(1) Only (b)

(2) (a) and (c)

(3) Only (a)

(4) (b) and (c)

161. All of the following can be the grounds for medical termination of pregnancy in India, **except**

(1) To get rid of unwanted pregnancy due to failure of contraceptive used during coitus

(2) When continuation of pregnancy could be fatal to the mother

(3) When no significant movement of foetus is yet observed by the end of first 8 weeks of pregnancy

(4) Due to the substantial risk that the child born would suffer from mental abnormalities

162. Introduction of sex education in schools should be encouraged to provide information to the young children

(1) To encourage them for believing in myths about sex-related aspects

(2) About reproductive organs and practicing unsafe and unhygienic sexual practices

(3) To make them not believe in misconceptions about adolescence and sex related changes

(4) To encourage them to be casual about STIs as they are always curable

163. Read the following statements w.r.t. humans:
Statement (A) : The mature sperm exhibits a remarkable degree of structural and functional specialization.
Statement (B) : The plasma membrane of sperm encloses the head and neck but not the middle piece and the tail.
Select the correct option.

(1) Both statements (A) and (B) are correct

(2) Both statements (A) and (B) are incorrect

(3) Only statement (A) is incorrect

(4) Only statement (B) is incorrect

164. Menopause in human females is most likely due to the

(1) Sudden decrease in the levels of GnRH

(2) Degeneration of ovarian follicles and decline in the levels of estrogen

(3) Increased production of ovarian hormones

(4) Increased amount of prolactin in plasma due to decrease in GnRH

165. Select the correct match w.r.t humans.

(1) Spermiogenesis – Spermatozoa are transformed into spermatids

(2) Spermiation – Release of spermatozoa into the seminiferous tubules

(3) Parturition – Initiated by decrease in estrogen and progesterone ratio

(4) Menstruation – Middle layer of uterus undergoes maximum changes

166. How many of the contraceptive devices mentioned below do(es) not require expert nurses/doctors for its/their insertion?

Condoms, Implants, LNG-20, Lippes loop, Diaphragm

Choose the correct option.

(1) Five

(2) Three

(3) One

(4) Two

167. As compared to the spermatozoa, the spermatids are/possess

(1) An altered nucleus and a tail

(2) Non-motile cells

(3) Small and diploid cells

(4) Cap-like structure called acrosome

168. Select the incorrect one w.r.t amniocentesis.

(1) It is a pre-natal diagnostic test

(2) Involves analysis of foetal cells present in amniotic fluid

(3) Completely banned in India

(4) Can determine the survivability of foetus by testing the presence of genetic disorders

169. The cell formed in the female gonads which possesses only one sex chromosome is

(1) Second polar body

(2) Primary oocyte

(3) First polar body

(4) Oogonium

170. **Assertion (A):** During pregnancy, the levels of hormones like estrogen, progesterone, thyroxine, cortisol, *etc.*, are increased several folds in the maternal blood.
Reason (R): In pregnant women, increased production of hormones is essential for supporting the foetal growth, metabolic changes in mother and maintenance of pregnancy.
In the light of above statements, choose the correct option.

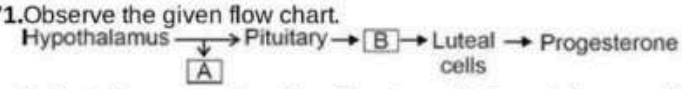
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) Both (A) and (R) are false.

171. Observe the given flow chart.



Select the correct option for A and B w.r.t hormonal regulation in human females.

(1) A = GH, B = FSH

(2) A = GnRH, B = Prolactin

(3) A = GnRH, B = LH

(4) A = GH, B = Estrogen

172. Choose the correct option to complete the analogy.
Viral STI : Hepatitis-B : : Protozoan STI : _____

(1) Genital warts

(2) Chlamydiasis

(3) Genital herpes

(4) Trichomoniasis

173. In a healthy adult female with 28 days of menstrual cycle, if the ovum is not fertilised then the corpus luteum starts to degenerate about

(1) 7 days before ovulation

(2) 4 days before next menstruation

(3) 4th day after ovulation

(4) 10 days before next menstruation

174. According to 2011 census report, our population is expanding rapidly. To check this growth rate, the Government of india has taken up measures such as
(a) Motivate small families to increase in size by using various conception methods.
(b) Statutory raising of marriageable age of females and males.
(c) Incentives given to couples with small families
Select the correct option.

(1) (a) and (b)

(2) Only (a)

(3) (b) and (c)

(4) Only (c)

175. The world population around the year 2000 was about _____times than the world population around year 1900.
Choose the correct option to fill in the blank.

(1) Three

(2) Seven

(3) Six

(4) Two

176. Which of the following hormones can delay menstruation if taken continuously throughout the menstrual cycle?

(1) Oxytocin

(2) Progesterone

(3) Estrogen

(4) Prolactin

CC-108 CC-108 CC-108 CC-108 CC-108 CC-108
Aakash Medical|IIT-JEE|Foundations

177. All of the following are incorrect w.r.t placenta, **except**

(1) Secretes luteinizing hormone

(2) Develops from the interdigitation of yolk sac and myometrium

(3) Allows direct mixing of maternal and foetal blood

(4) Formed by chorionic villi on foetal side

178. Voluntary termination of pregnancy is risky at 22nd week of pregnancy, thus, it is

(1) Not recommended in any condition

(2) Done by unqualified quacks legally

(3) Done after mandatory recommendation of 4 medical experts

(4) Done after opinion of 2 registered medical practitioners

179. A proteinaceous female gonadal hormone

(1) Interacts with intracellular receptors

(2) Has similar mechanism of action as testosterone

(3) Is secreted in the later phase of pregnancy to facilitate the delivery of foetus

(4) Constricts the cervix and thickens the cervical mucus

180. If a human male is unable to inseminate his female partner due to erectile dysfunctioning, then which of the following ARTs should be recommended to the couple that also involves *in-vitro* fertilization?

(1) GIFT

(2) AI

(3) IUI

(4) ICSI

181. Select the odd one w.r.t features of androgen in males.

(1) Stimulates gametogenesis

(2) LH stimulates its secretion

(3) It increases the pitch of the voice

(4) It is secreted by interstitial cells of testis

182. To attain reproductive health as a social goal, India first initiated the action plans and programmes at a national level in the year

(1) 1951

(2) 1971

(3) 2000

(4) 2011

183. An ideal contraceptive should not

(1) Be user-friendly

(2) Be easily available to the user

(3) Be reversible

(4) Interfere with the sexual drive of user

184. Read the following statements

Statement (A) : A female condom is a barrier contraceptive device that provides protection against STIs and constitutes two rings at both the ends, one of which is open unlike the other.

Statement (B) : All barrier contraceptives provide privacy and are reusable.

Select the correct option.

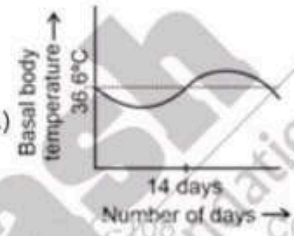
(1) Both the statements (A) and (B) are correct

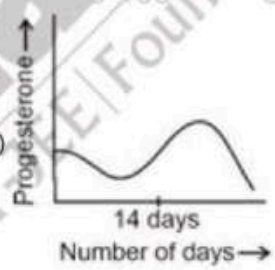
(2) Both the statements (A) and (B) are incorrect

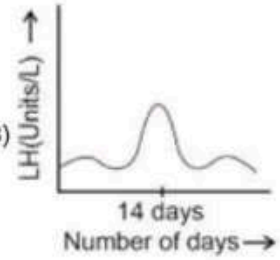
(3) Only statement (A) is correct

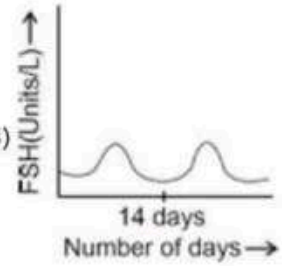
(4) Only statement (B) is correct

185. Which of the following graphs correctly depicts one of the causes of ovulation in human females?

(1)


(2)


(3)


(4)


CC-108 CC-108 CC-108 CC-108 CC-108 CC-108 CC-108

SECTION-B

186. During fusion of a human sperm with an ovum in ampulla region of fallopian tube, the sperm firstly comes in contact with

(1) Zona pellucida

(2) A cellular layer

(3) An acellular layer

(4) Basement membrane

187. Lactational amenorrhea is based on the fact that

(1) Increased hPL inhibits the secretion of TSH

(2) Decreased suckling action provides negative feedback to GnRH

(3) Increased PRL inhibits the secretion of GnRH

(4) Increased secretion of oxytocin from anterior pituitary inhibits GnRH secretion

188. **Assertion (A):** Levonorgestrel-20 is a contraceptive which makes the uterus unsuitable for implantation and the cervix hostile to sperms.
Reason (R): It contains synthetic progesterone which causes thinning of cervical mucus.
In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(2) Both (A) and (R) are false.

(3) (A) is true but (R) is false.

(4) Both (A) and (R) are true and (R) is the correct explanation of (A).

189. Match the column I with column II

	Column I		Column II
a.	Syphilis	(i)	Protozoan STI
b.	IUT	(ii)	*In-vivo* fertilisation
c.	IUI	(iii)	*In-vitro* fertilisation
d.	Genital warts	(iv)	Bacterial STI
		(v)	Viral STI

Select the correct option

(1) a(v), b(ii), c(iii), d(i)

(2) a(v), b(iii), c(ii), d(iv)

(3) a(iv), b(ii), c(iii), d(i)

(4) a(iv), b(iii), c(ii), d(v)

190. Multiload 375 is

(1) Inserted into the vagina and is fixed at the opening of cervix

(2) A non-medicated IUD

(3) Placed under the skin, where it secretes progestogen at intervals

(4) Capable of releasing Cu ions which alters the fertilising capacity of sperms

191. During parturition, release of oxytocin and strength of uterine contractions exhibit

(1) Positive feedback loop

(2) Negative feedback loop

(3) Both negative feedback and positive feedback loop

(4) Neither negative feedback nor positive feedback loop

192. The mammary glands of human females can undergo structural differentiation under the effect of steroidal hormones
(a) Progesterone
(b) Estrogen
(c) Oxytocin
Select the correct option.

(1) (a), (b) and (c)

(2) Only (a) and (c)

(3) (a) and (b)

(4) Only (b) and (c)

193. Which of the following will not occur after vasectomy?

(1) Transport of sperm from testis to ejaculatory duct

(2) Presence of sperms in epididymis

(3) Production of androgens

(4) Spermiogenesis

194. Which among the following is more susceptible to HIV infection?

(1) One who avoids sex with unknown partners

(2) One who uses condoms during coitus

(3) One who uses Saheli during coitus

(4) One who does not share injection needles with others

CC-108 CC-108 CC-108 CC-108 CC-108 CC-108 CC-108
Aakash Medical|IIT-JEE|Foundations

195. Contraceptives are

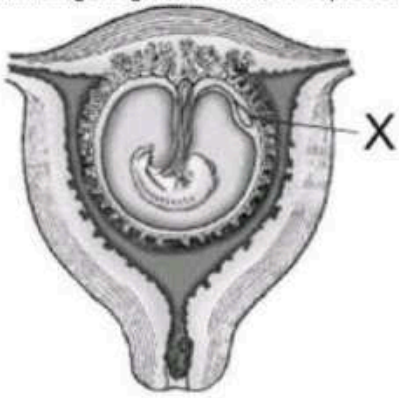
(1) Regular requirements for maintenance of reproductive health

(2) Capable of causing breast cancer in some cases, if taken for a prolonged period

(3) Neither used to prevent pregnancy nor to delay pregnancy

(4) Used widely to promote uncontrolled growth of population

196. In the figure given below, 'X' represents



(1) Placental villi

(2) Embryo

(3) Yolk sac

(4) Umbilical cord

197. Read the following statements

(a) Implantation usually occurs when embryo contains 8-16 blastomeres.

(b) Sickle-cell anaemia, haemophilia and down syndrome are genetic disorders that can be detected *via* amniocentesis. .

(c) Indian population reached to the billion mark by the time of independence.

(d) Ectopic pregnancy and still birth can occur due to untreated STIs.

How many of the above given statements is/are correct?

(1) Four

(2) Two

(3) Three

(4) One

198. Select the incorrect option w.r.t uterus of human female.

(1) Inverted pear-shaped

(2) Present in pelvis between bladder and rectum

(3) Outer most layer of its wall is thin and membranous

(4) Fundus is its lower narrow part

199. In humans, secondary follicle and tertiary follicle are present in

(1) Inner medulla of ovarian stroma

(2) Isthmus of oviducts

(3) Cortex region of ovarian stroma

(4) Fundus of uterus

200. Select the incorrect **option** w.r.t. sex accessory glands present in human males.

(1) Secrete seminal plasma which is rich in fructose, calcium and certain enzymes

(2) Secretions of bulbourethral glands help in the lubrication of penis

(3) Seminal vesicle and prostate are paired glands

(4) Seminal plasma along with sperms constitute semen

CC-108 CC-108 CC-108 CC-108 CC-108 CC-108 CC-108